

some stands. The sites of some others are indicated by the nature of present-day relics. Tawa is shown in a number of places on Stewart's maps but presumably these are individual trees marked during the survey. However, in three places tawa forest is indicated. "Tawa trees" is marked within a loop of the Manawatu River on the site of Poplar Rd. Between Te Waiti swamp and the Oroua River there is a reference to "tawa bush" and "tawa and small timber" along the survey line. A further stand of tawa is indicated on the southern margin of Papaioea clearing. The tawa forests indicated on the maps are all near survey lines. It is unsafe to guess at what other parts of the plains may have carried tawa forest. The present-day remnants suggest that there was tawa forest on the hills and terraces east of the Manawatu River.

THE PRESENT

Distribution

Most of the tawa forest which has survived is on the hills at the Manawatu Gorge and in the Tiritea catchment. From the tawa forest in the State Forest at the gorge numerous remnants in gullies extend south-westwards to meet the beech forest near Forest Hill Rd. A larger tract of tawa forest remains in the Tiritea valley almost wholly within the boundaries of the water reserve (figure 15). It extends from the region of the dams along the Little Tiritea Stream almost to the top of the range, and for a similar distance along the upper reaches of the main stream and its tributaries. Outside the reserve there is an isolated patch skirted by Greens Rd, and numerous other remnants between the Tiritea and Kahuterawa Streams. Pockets of tawa in other places are most likely to be degenerate podocarp-broadleaf forest.

Composition and structure

In the State Forest at the Manawatu Gorge tawa forms a pure canopy in a few pockets which have probably been logged for podocarps but not burnt. The tawa trees there are very large, reaching over 30 m, with very little understorey apart from a few mahoe and nikau palms which are suffering from the low illumination under the closed tawa canopy.

Tawa forest within the Tiritea reserve is the best-preserved forest of any kind in the Manawatu. In spite of the impact of deer, cattle, sheep, and pigs, tawa has been able to remain in a healthy condition. A fair degree of protection from the strong prevailing winds has undoubtedly played a part in its survival. Except for the demise of a few northern rata and kamahi this stand is probably still typical of the forest of a century ago. A few old scattered podocarps, mostly rimu, stand above an even canopy composed almost solely of tawa. The tallest tawa trees reach over 24 m and have a diameter of



Figure 15 The best preserved tawa forest is in the Tiritea reserve. A few old emergent rimu and northern rata remain. 1966

about 1 m. Most are shorter than this and small-diameter classes predominate. Mahoe and pigeonwood are the most common associated species. Others frequently found in tawa forest are hinau, rewarewa, and toro.

On the floor of the tawa forest there is an abundance of *Blechnum filiforme* and *Ctenitis glabella*. The stems of *Metrosideros diffusa* and *M. perforata* scramble across the stumps and here and there climb up towards the canopy. There is a considerable amount of *Asplenium bulbiferum* considering the severity with which it has been browsed. *Microlaena avenacea* and *Uncinia uncinata* are abundant only where the canopy is broken. *Blechnum discolor*, so abundant in other parts of the water reserve, occurs occasionally in the tawa forest. The shrub layer is fairly open except in the damp places where there are tangles of supplejack and kiekie. The main species in the shrub layer are tawa saplings, kawakawa, *Geniostoma*, and *Coprosma rhamnoides* all overtopped by *Dicksonia squarrosa* and three species of *Cyathea*. *Mida* and *Rhabdothamnus* are associated more frequently with tawa forest than any other type. *Rhabdothamnus* grows on steep moist slopes near streams. Nikau palms are present in small numbers. Throughout the tawa forest there are numerous

stumps of podocarps which have reached the end of their lives and of northern rata trees which have had their lives shortened by opossum browsing.

Beech forest

THE PAST

Distribution

Pockets of black beech occur in many places in the courses of the Manawatu and Pohangina Rivers but there is only one in the region under study. From a survey of the distribution of stumps it is estimated that beech forest covered about 526 ha in the Aokautere district. It extended about 3.2 km north-east of Forest Hill Rd and about half this distance to the south-west, and in altitude from 90 to 275 m, just crossing the boundary of the terrace country onto the greywacke.

THE PRESENT

Distribution

Hard beech grows in the Aokautere area but is quite rare. There are a few trees, and hybrids with black beech at 213 m in a forest remnant in a branch of Te Taupiri Stream.

F. G. Hayes (pers. comm.) recalls seeing a few beech poles at the junction of South Range Rd and Pahiata Track about 1927 but this sighting cannot be confirmed as the land is now in pasture. I do not doubt the accuracy of his observations.

Composition and structure

The most striking feature of the beech forest remnants is the dominance of beech and the abundance of *Cyatodes fasciculata* and *Coprosma rhamnoides*. The beech trees are in the 12–18 m height range and few exceed 0.5 m in diameter. They are quite young and were never part of the canopy of the original forest. The scrubby understorey is doubtlessly animal-induced. The gross disturbance, the small size of the remnants and the indistinct boundary with other types of forest make it impossible to state the composition of the primary forest. Three shrubs, *Cyatodes fasciculata*, *Helichrysum glomeratum*, and *Gaultheria antipoda*, are more abundant on beech forest sites but this cannot be taken to mean that they had a significant place in the primary forest.

Northern rata – kamahi forest

THE PAST

Distribution

The extent of northern rata – kamahi forest is not known for certain but

forest of this type seems to have occupied most of the hill country except where tawa forest existed. Records of kamahi are difficult to obtain as most settlers seem to have overlooked the species or did not identify it. The tree species of tall stature, and those producing timber, were well known but kamahi was regarded as "one of the others". A number of local common names may also have contributed to this vagueness. Tracts of northern rata – kamahi forest were regarded as poor timber country. Perhaps it was this attitude that prompted J. T. Stewart to write on his map (S.O. 10603) "mostly inferior timber" in the region of the middle section of Scotts Rd.

The vegetation of relics of the peneplain e.g., around Ramiha trig, where the soils are deeper and contain andesitic ash, may also have been northern rata – kamahi forest. Most of the soils of this origin have been sown to pasture and where native vegetation remains it is so badly damaged that no useful information can be gleaned.

Composition and structure

Fortunately, my association with the water reserves began in 1959 when there was enough remaining of the primitive vegetation to enable the mental reconstruction of the early forest. I was aided in this task by the very clear recollections of F. G. Hayes who trapped opossums in the region in 1927 before the major part of the destruction became obvious.

The picture of the primitive forest that emerges is a multi-tiered forest with a considerable amount of northern rata, miro, and rimu above a subordinate layer of kamahi, tawa, and black maire with conspicuous amounts of *Griselinia littoralis* and Hall's totara above 365 m and an abundance of tawa below this altitude (figure 16).

THE PRESENT

Distribution

A large proportion of the area occupied by northern rata – kamahi forest on the upper flanks of the Tararua Range a century ago is still covered in native vegetation. Much of the area is reserved as water catchments in the Kahuterawa and Tiritea valleys and a small portion outside the reserve is too steep to be farmed.

Evidence of vegetation of the past

Numerous disasters have befallen the area and the vegetation as a consequence is grossly degraded (figure 17). The vegetation is no longer northern rata – kamahi forest but a mosaic of shrubland and grassland. There was some evidence to suggest that a Maori fire swept part of the area about 1760 (Esler 1963) but subsequent devastation has obliterated the boundaries of the affected area. In 1898 a bush fire starting on the western side of the range

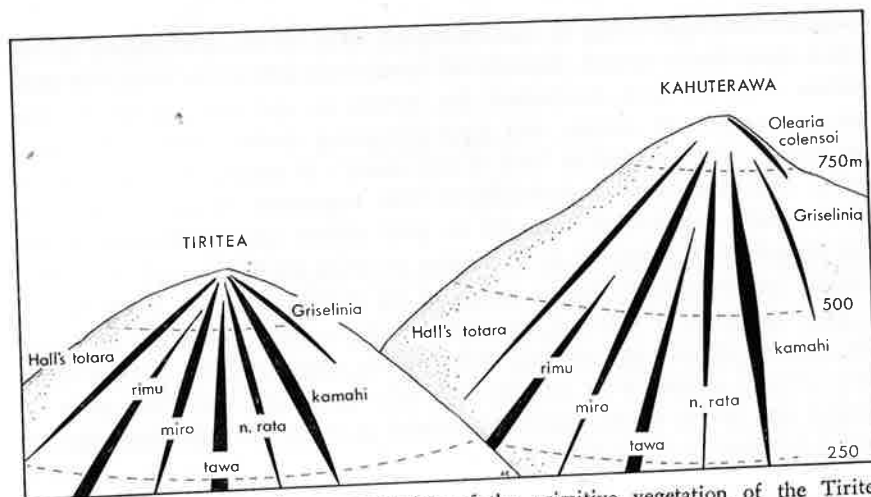


Figure 16. Diagrammatic representation of the primitive vegetation of the Tiritea and Kahuterawa catchments. Width of bar expresses estimated relative abundance based mainly on presence of stumps and living trees identified in the 1960s. Altitude is indicated in metres.

swept through a large part of northern Wairarapa. It is known that it ran near the water catchment but there the detail ceases. It is certain that fire did enter the reserve at other times, survey lines being no barrier to flames when the land was being cleared for farming. At various times severe gales have taken their toll. One of the most damaging gales struck in 1936. Deer, pigs, sheep, and cattle have had access to the reserves for many decades but it is to opossums that the most serious initial damage can be attributed.

The gaunt skeletons of northern rata (figure 18) in the Tiritea reserve when viewed from Palmerston North were a conspicuous feature of the skyline in the 1960s but towards the end of the decade most had fallen. From the distribution of the logs it appears that northern rata occurred fairly evenly throughout the reserve but was less abundant in the tawa-dominant areas. Trees over 30 m high stood on the crest of the range in places which now appear hostile to forest and carry only closely-cropped turf (figure 19). Until 1927 or later trees clothed the summit. Hayes wrote: "At that time, though the forest was open and the floor clean, there was a good canopy overhead. The only places the Manawatu plains could be seen from, were Marima trig itself . . . and from a couple of points on the 'Backbone' (south of Marima)". There are not more than a dozen old living northern rata trees in the reserve today.

Kamahi stumps in 1959 indicated the abundance of the species in the primitive forest. Like northern rata it was less plentiful where tawa was abundant. When I first became acquainted with the Tiritea the stumps,



Figure 17 Degraded northern rata-kamahi forest in the Otangaue catchment. Opossums were responsible for much of this damage. 1960

many of them more than 2 m in diameter, were in an advanced state of decay but, in a manner characteristic of kamahi, retained bark which provided ready identification of the species. All stumps were festooned with filmy ferns. By 1970 most of the evidence of the former abundance of kamahi had disappeared except in some restricted areas where patches of kamahi remained alive. The pattern of deterioration was observed in a stand of kamahi, tawa, and rimu which was in a vigorous state in 1959. Thereafter the kamahi took on a greyish appearance as it was defoliated by opossums. The defoliation was closely examined and there was not the slightest doubt about its cause. Death followed within a year or two and by 1970 most of the limbs had fallen. Seedlings of kamahi occur on roadside banks near the lower dam and as epiphytes in a few places but regeneration is insignificant.

Rimu and miro once occurred in all parts of the reserve and logs of podocarps, presumably these two species, are prominent in many places. At present there



Figure 18 Skeletons of northern rata were a feature of the Tiritea skyline in the 1960s. 1964

are about equal numbers of rimu and miro in the gullies but in the more exposed places miro outnumber rimu. It is not known if this reflects their relative abundance in the primary forest. Both have straight, clean boles indicating that they developed with other trees of tall stature. Matai are few in number.

Decaying logs indicate that black maire, Hall's totara, totara, and *Griselinia littoralis* were all more abundant than at present. Black maire was quite plentiful but is now a rare species. Hall's totara logs appear mostly at higher altitudes but it is evident from the presence of a few living trees in the tawa forest that they were at lower levels also. Evidence of *Podocarpus totara* is mainly in the north and west of the reserve but the species was not confined to these places. Cut stumps near the boundaries indicate the fate of many of them. *Griselinia littoralis* was probably associated with Hall's totara in fairly high numbers above 365 m but the only plants existing today are of a

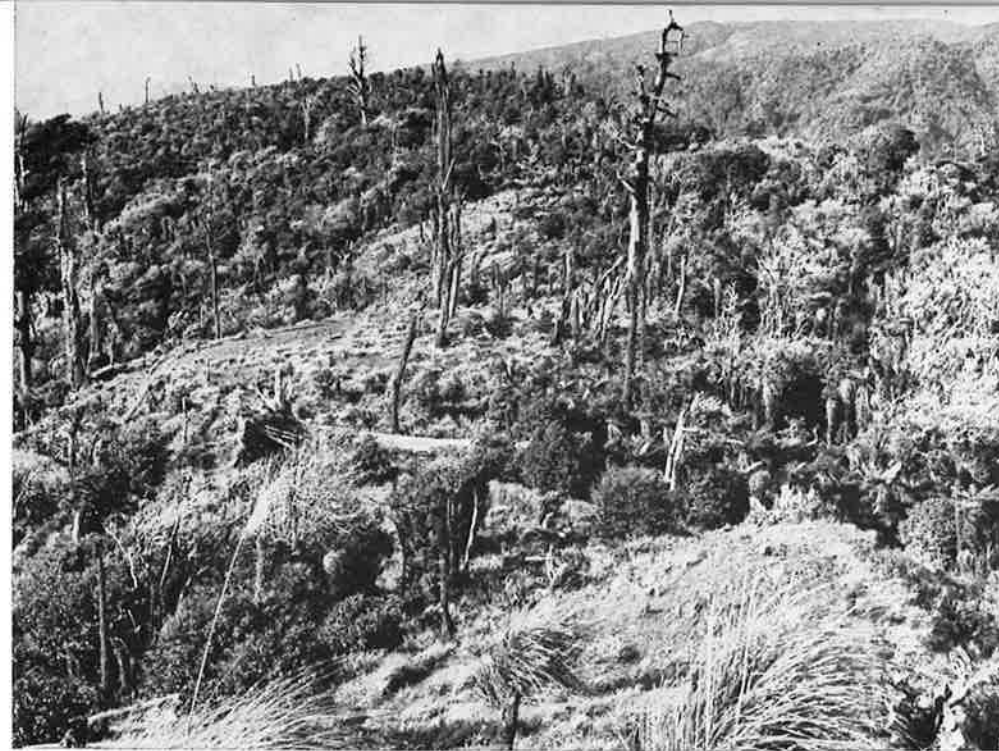


Figure 19 Summit ridge in the Tiritea reserve. In 1927 this ridge was densely clad in forest and there were no view points onto surrounding country. *Cortaderia toetoe* grows in the foreground. 1964

later epiphytic generation and have their foliage above the browse level of deer (figure 25).

Several other species with shorter life spans, and mainly of secondary forest, probably had a place in the primary forest but their significance is unknown. These include mahoe, hinau, rewarewa, and toro.

The nature of the understorey of the northern rata-kamahi forest is unrecorded because it had disappeared before 1927. Hayes recalls: "Forest was very open. I do not remember cutting a single stick except for firewood and to build my camp. I never carried an axe or slasher (only a light tomahawk). Floor of the forest was very clean. No seedlings or young trees except in odd small patches, little fern and other damp-loving vegetation. No pig or goat sign, but deer were numerous." At the present day most of the canopy is too low to have an understorey.

In the Kahuterawa valley the northern rata-kamahi forest appears to have been similar to that of the Tiritea (figure 16). The summit of the range being 275 m higher allowed the expansion of miro, kamahi, northern rata,

and Hall's totara to a higher altitude where they ran into a small stand of *Olearia colensoi*. Hall's totara may have been more abundant here than in the Tiritea valley and kamahi logs at lower altitudes seemed more numerous. There was vigorous kamahi regeneration in 1959 in the region of Kahuterawa Flat but a later wave of opossum destruction greatly reduced the amount of kamahi. From the state of degradation of the vegetation and the decay of the logs it seems that the period of mass destruction of the canopy followed about 10 years after the devastation in the Tiritea catchment, but there is no more evidence than this from which to draw conclusions.

Forests of the dunes

THE PAST

Distribution

Cockayne (1928, page 95), in his survey of the vegetation of New Zealand, stated simply: "Forest is rare on the dune-areas". Some such forest existed in the Manawatu. Early plans show about 2100 ha of forest in patches on the sand country near its eastern margin from Carnarvon to Moutoa but approaching to within 4.8 km of the coast in the south. Only some of this would have been dune forest, the remainder being semi-swamp forest on the intervening plains.

Comparing forest distribution with the soil map we find that the forest patches are associated mainly with the third dune phase which is estimated by J. D. Cowie (pers. comm.) to have been stable for 5 000 years. The dunes of this phase are low, mostly less than 9 m high and their topsoils more than 25 cm deep and black (Foxton black sand) or less commonly brown (Foxton brown sand). Foxton black sand is stated (Cowie and Smith 1958, page 22) to have developed under bracken and manuka and the Foxton brown sand under forest (Cowie *et al.* 1967, page 17). The maps of soil and vegetation bear this out in a general way but the distinction is not clear cut.

Ecologists may ask why forests did not blanket the whole of the dunes of the third phase because the habitat does not seem to be adverse to forest. No answer can be provided, but it must be borne in mind that forest could have been destroyed many centuries before the Europeans arrived.

Composition and structure

No very clear picture of the primitive dune forest can be gained from the minor relics which remain. Probably the dominant trees were mahoe, titoki, *Dodonaea*, and ngaio, with tawa prominent in parts. There may also have been some totara.

THE PRESENT

Distribution

The forested area has now contracted to about 4 percent of its former extent and by good fortune some of the dune forest, which was only a minor part of the whole, has survived. Some of this is protected in Himatangi Bush Scientific Reserve, a relic which would certainly have soon disappeared had this survey of the vegetation of the Manawatu not been undertaken.

The greatest extent of the remaining dune forest is at the boundary between the second and third dune phases where the high second phase dune spills over the lower older dunes. These high dunes have dark grey topsoils (Foxton dark grey sand) not more than 15 cm deep. The presence of forest on these soils must be viewed with caution as it is obvious in some places that sand from the second phase has spread over older soils without destroying the trees. At Whites Bush, for instance, kahikatea can be found growing on second phase dunes and at Himatangi Bush *Eugenia maire* is growing on a dune. While this has occurred in some places it is quite clear that dunes of the second phase have supported forest in parts.

At Himatangi Bush the forest extends onto dunes with deep black topsoils but the area is so limited and the wet soils of the sand plains so close that it is unsafe to state that the mahoe, cabbage tree, tawa, and *Melicope simplex* are characteristic species of the dunes of the third dune phase. I am unaware of any remnants of forest on Foxton brown sand.

Composition and structure

At Himatangi Bush the semi-swamp forest occupying the old and wet soils of the sand plains gives way to tawa on the edge of the dune. This narrow belt might appear to owe its existence to the high water table but tawa continues less densely to near the top of the 18 m dunes.

The impression is gained that dune forest has little to distinguish it floristically from other remnants of broadleaf forest in other parts of the Manawatu. No one species characterises the dune forest but the combination of species does not occur elsewhere in the district.

The physiognomic dominant trees are tawa, mahoe, titoki, *Dodonaea viscosa*, and ngaio but there are appreciable quantities of rewarewa, mapou, kaikomako, hinau, lancewood, pigeonwood, kowhai, *Pittosporum tenuifolium*, and kanuka. The *Dodonaea* and ngaio may be considered to reflect the coastal influence but both grow in the Manawatu Gorge 40 km from the coast and ngaio has been seen on greywacke near Forest Hill Rd about 275 m above sea level.

The understorey has doubtlessly been modified by cattle browsing out palatable species and eroding the steep sides of the dunes. It contains *Coprosma*

rbamnoides, *C. crassifolia*, *Cyatodes fasciculata*, *Melicope simplex*, *Carmichaelia flagelliformis*, *Rubus squarrosus*, *Parsonsia capsularis*, *Clematis forsteri*, and *Metrosideros colensoi*.

The ground flora is surprisingly rich considering the amount of soil disturbance caused by cattle. In the most disturbed areas *Parietaria debilis* abounds and this is soon joined by *Oplismenus* to give very rapid stabilisation. The same habitat provides a place for *Arthropodium candidum*, *Microlaena stipoides*, *Echinopogon ovatus*, *Corybas trilobus*, *Pratia angulata* and the ferns *Polystichum richardii*, *Pteris macilenta*, *P. tremula*, and *Pellaea rotundifolia*.

At Whites Bush the forested dune is very sparsely populated with trees because of the stage of deterioration. Here there is doubt about the age of the sand around the trees. The topsoil is very much younger than third dune phase but this may just be a superficial and recent layer. It is certain that sand was deposited around the bases of tawa and kahikatea on the lee face of the dune. The few large totara trees appear to be occupying second-phase dune. However, totara can tolerate the accumulation of sand above its roots so it is reasonable to have some doubts. On the other hand the fact that these trees are on the windward side of the dune lessens the probability of sand having built up around them after they established.

There is ample evidence on the dunes near Himatangi Bush to show that a wide range of forest species can establish successfully high on a dune away from the direct influence of the water table of the plains. Pre-forest shrubs such as lancewood, mahoe, five-finger, wineberry, *Pittosporum tenuifolium*, and cabbage tree are establishing freely in bracken and kanuka on both sunny and shaded aspects but they are more abundant on slopes lying to the south. An isolated relic of forest on the crest of a dune near Pukehinau trig (grid ref. N148/845222) is evidence of the ability of forest to develop on Foxton dark grey sand away from large tracts of forest. This remnant caps the dune and lies partly in a trough along its summit. Ninety percent of the trees are titoki, the remainder of the canopy being composed of mahoe, ngaio, kanuka, and cabbage trees. The only other significant native plants are *Rubus schmidelioides*, *Muehlenbeckia complexa*, and a possible hybrid between *M. complexa* and *M. australis*. The maturity of the stand suggests that a well-developed flora could have been maintained but intensive trampling and erosion have taken their toll.

The presence of forest on some of the dunes of the second dune phase raises the perplexing problem of the nature of the primitive vegetation of the sand plains of the same age. The best information that we have suggests that vegetation on these developed only as far as shrubland. This is anomalous. The plains have a much more abundant supply of moisture, a critical factor in a low-rainfall region with only 812 mm per annum, and are much less

exposed to the strong onshore winds and the salt that they bear. I believe that in the absence of fire, and given a source of seed, forest of some sort could have developed on many parts of the sand plains within 500 years of stabilisation of the sand. I do not rule out the possibility of suppression by fire.

SWAMPS AND OTHER WETLANDS

Swamps

THE PAST

Distribution

Unless swamps are bounded by higher ground they often merge into surrounding vegetation and have ill-defined margins. Where the true margin lies can be a matter of opinion. For this reason boundaries of swamps are not always clearly marked on maps and plans. Because of this difficulty, and because all swampy areas were not mapped, it is not possible now to measure accurately the extent of swamps of a century ago. My estimate puts their area in excess of 8000 ha or over 8% of the land surface of the Manawatu.

The main swamps shown on the early maps and plans are the Moutoa, Makerua, and other swamps along the Manawatu and Oroua Rivers, the Taonui basin and those associated with the coastal lagoons.

Composition and structure

Although swamps were a major feature of the landscape and held an important place in the economy of the early settlement little detail of their botany can be found. Passing comments were made by Kettle (1842) and Wakefield (1845) in the 1840s and by early surveyors in the following decade. The Moutoa Swamp was marked by Stewart (S.O. 10639) thus: "The Great Swamp. Flax, raupo, &c. deep soil with imbedded timber". It is apparent from the map that there were lagoons also, and fringes of forest on the levées near the Manawatu River.

Kettle (1842) viewed the Makerua Swamp from near the river end of Tane Rd. He commented, ". . . but at present is quite dry and not of a swampy nature as is the generality of the toi toi land about this part of the country". It is said of this swamp, as of the Moutoa Swamp, that *Phormium* increased following drainage. This implies that *Cortaderia* (presumably *C. toetoe*) occupied wetter sites than *Phormium*. It is recognised that *C. toetoe* can grow in very wet soil but there is now insufficient undisturbed swamp to verify this point.

The plan by Knowles (S.O. 10837) shows the limits of the head of the Makerua swamp but his annotations (omitted from the 1959 redraft) and his field book add little detail of the vegetation. Stewart's map (S.O. 10639) bears the words: "Hard bottom at 8 to 10 feet under surface, a pole being easily pushed to that depth". Nearby is the note: "flax, toe toe, raupo, rushes &c.". Since development for farming the surface has settled a metre or more and stumps protrude through the soil. These are relics of an extensive forest which clothed the river flats before forests gave way to swamp.

A small swamp in the Glen Oroua district was traversed by Wakefield (1845, volume 2, page 235) in his walk from Parewanui near the Rangitikei River to the Oroua River. This was probably about 3 km south of Rongotea where a clearing is shown on early plans of the Douglas block. He wrote: "Towards dusk we entered a spacious kind of bay among the wood, and reached the borders of a swamp which filled one-half of it . . . At break of day, we followed a track across the swamp, about a mile, to the edge of the wood. The water was only a little above our knees after all . . .". *Phormium* grew in the swamp but little other detail is recorded.

Between Feilding and Kopane, Stewart marked "Open flat. Flax, toe toe &c." in one place and "Open flat. Fern, flax, toe toe" in another.

The swampy areas between Rangiotu and Moutoa were obviously extensive but no plan shows any detail and there appear to be no recorded descriptions. There were numerous flax mills in the region taking leaf from both sides of the river.

Across Taonui Swamp Stewart (S.O. 10603) marked "Coarse grass in tufts. Dry in summer". A copy of this plan by Igglesden (S.O. 10604) bears the note: "Open flat, mostly dry in summer. Coarse grass, docks &c.". The justification for the addition of "docks" is not clear!

It seems that there are no recorded details of the nature of the swamps around the coastal lagoons.

THE PRESENT

Distribution

Very little remains of the swamps of the Manawatu except for limited areas near the coast.

Composition and structure

Poole and Boyce (1949) detailed part of the Moutoa Swamp after some modification had taken place. The swamp lies only a few metres above sea level and the river rises and falls with the tide. Before stopbanks were erected the swamp was inundated several times each year with flood water. At the time of the survey by Poole and Boyce artificial drainage had altered the pattern of distribution of plant communities but they were able to

recognise the zonation that existed before modification. In an elevation of about 1.8 m from the bottom of the lagoons to the summit of the levées the vegetation changed from *Carex secta* in the free water to raupo on the edges to *Phormium* to swamp scrub to semi-swamp forest. The swamp scrub was stated to consist of *Cordyline australis* and *Coprosma* spp. The list contains *Coprosma areolata*, *C. propinqua*, and *C. robusta* for the "semi-swamp" habitat. The semi-swamp forest was dominated by kahikatea and pukatea. Poole and Boyce listed 66 native species recorded on the river bank, 55 of them in forest. The semi-swamp flora contained about 40 native species. In a description of the *Phormium* stands they listed the following native plants—*Blechnum capense*, *Carex ternaria* (probably the entity now known as *C. lessoniana*), *C. secta*, raupo, *Coprosma propinqua*, *C. robusta*, *C. propinqua* × *robusta*, *C. areolata*, *Muehlenbeckia australis*, *M. complexa*, *Pittosporum tenuifolium*, and *Cordyline australis*. Such stands became very extensive after drainage and stopbanking. This encouraged the spread of *Phormium* over the land previously too wet for the species.

Along the Manawatu coast sand dunes moving inland formed a barrier to the movement of water from the gently sloping hinterland and a string of lagoons has formed a little more than 3 km from the beach. Extensive swamps were associated with each. Other lagoons of this nature have occupied the same country in the past. In the region of Round Bush, for instance, there is a compressed band of peat 3.5 to 4.5 m below the surface (J. W. Barber pers. comm.). The displacement of Pukepuke Lagoon and Lake Koputara by drifting sand since the first plans were drawn illustrates the manner in which these disappeared. The site of Pukepuke on the early survey plans is now pasture land.

It is unfortunate that the early plans do not show the extent of the swamps around the coastal lagoons more fully. The swampy land around Koputara is shown clearly (S.O. 10987) because the lagoons were part of a native reserve being surveyed but the southern group of lagoons have been omitted from this and other plans available to me. Kaikokopu Lagoon is shown in part on this same plan and swampy land indicated at the north-eastern extremity. An interesting 1867 plan reproduced by Downes (1909, plate xi) shows the Kaikokopu lagoons and extensive swamp surrounding them. This swamp was apparently continuous with the swamp associated with Pukepuke Lagoon.

The vegetation of the coastal lagoons, as a consequence of the mobility of the sand, is young. It has been modified by manipulation of the water table, grazing and trampling by cattle, and by the ingress of alien plants. In spite of these changes few, if any, plant species have been lost. However, it seems likely that the distribution and composition of the plant communities have been influenced to a major extent.

The striking feature of the swamps is the abundance of raupo, *Phormium*, and cabbage trees. The same three species probably dominated during last century but raupo has spread very rapidly in recent times. This is revealed in aerial photographs of Pukepuke Lagoon taken by the RNZAF from Ohakea and by Lands and Survey Department at numerous times in the last 30 years. Lowering of the water levels seems to have been a contributing cause. The deepest part of Pukepuke Lagoon (figure 20) is no more than 0.6 m deep and raupo spreads in a raft-like mass from the margins into the open water.

In the few places not yet dominated by raupo, some pattern of zonation is discernible. There is a sequence beginning in open water with *Scirpus lacustris*, *Potamogeton crispus*, *P. pectinatus*, *Ruppia polycarpa*, *Ranunculus fluitans*, and floating plants of *Azolla rubra*, *Lemna minor*, *Wolffia arrhiza*, and *Spirodela oligorrhiza*. Near, and on, the muddy margins there is a band of *Nasturtium officinale*, *Veronica anagallis-aquatica*, *Ranunculus sceleratus*, and *Callitriche stagnalis*. Beyond this there is a sward of *Juncus articulatus* and *Agrostis stolonifera* merging into *Carex secta*. There is some evidence of *Carex secta* standing in open water as Poole and Boyce (1949) stated for Moutoa but

Figure 20 Pukepuke Lagoon. The prominent plants are raupo and *Carex secta*. 1969



the plants tend to be more numerous near the outer fringe. They may have stood in water when the lake level was higher. In Lake Horowhenua, Levin, *Carex secta* was left high and dry when the lake was lowered but continued in a healthy state till the clumps were destroyed by fire to reclaim more land for farming.

Phormium and cabbage trees tend to occupy the higher ground but most of this land around the lagoons has been brought into pasture. Where cattle have had access to the *Phormium* the plants are stunted by the root damage caused by the trampling. This weakens the stands and prepares the way for the invasion by tall fescue, *Lotus pedunculatus*, *Juncus articulatus*, *Agrostis stolonifera*, and *Polygonum hydropiper*.

The raupo stands support the pink-flowered *Calystegia sepium* even in the densest parts but where the stand is more open (and possibly drier) there is *Phormium*, cabbage tree, *Cortaderia toetoe*, *Solanum dulcamara*, *Hypolepis tenuifolia* (fine form), *Carex secta*, *Bidens frondosa*, *Blechnum capense*, and *Urtica linearifolia*, this latter usually on the pedestals of *Carex secta*. The place of *Cyperus ustulatus* along the soil moisture gradient cannot be stated for certain as it has recently diminished in abundance and is now not a conspicuous part of wetland communities in any part of the Manawatu.

Estuarine wetlands

Coastal Manawatu is bounded in the north and south by estuaries. The Rangitikei River near its mouth runs near the southern bank and no wetlands fall within the area surveyed. The Manawatu River forms a large estuary with wetland vegetation on the margins and towards the main channel on mud exposed at low tide.

It is apparent that marked vegetation and physiographic changes have occurred in the Manawatu estuary since H. H. Allan made his observations in 1924. In his survey of the spread of *Spartina* Allan (1924a, page 256) gave a brief description of the salt swamps at Foxton. He mentioned that they were typical of those described by Cockayne (1921b) in his first edition of "The Vegetation of New Zealand" and went on to say: "*Juncus maritimus* var. *australiensis* and *Leptocarpus simplex* are dominants, both forming very extensive dense sheets, the latter usually seaward of the former, and less in extent. There is not wanting evidence that in this locality the *Juncus* invades the *Leptocarpus* sheets and tends to subdue them. The areas of bare mud are not extensive, and into the *Spartina* neighbourhood border the tidal channels. Between the margins of the dominants and the bare mud there may be patches of *Scirpus americanus*, *Salicornia australis*, or *Triglochin striata* var. *filifolia*, the latter pair with or without other small species". Some later comments are given by Allan (1930).

Now, nearly half a century later, the same species remain in the estuary but their distribution and abundance have vastly altered. Vegetation of this type is restricted to a few minor places on margins of the estuary and much bare mud is exposed. Since 1924 the course of the lower Manawatu River has been shortened by cutting off the Foxton loop and forming the Moutoa floodway. An old resident told me that when the flood gates were opened several years ago the flood water took away "30 to 50 acres of sand" from the south head at the mouth of the river. This exposed the estuary to the sea and the areas formerly silty became sandy. It is not clear if this change in the texture of the substratum or erosion and accretion destroyed the large areas of salt swamp mentioned by Allan (1924a) and shown in his illustrations. (Note that the captions to his figure 1 and figure 2 are transposed).

Juncus acutus, one of the distinctive exotic rushes of the Foxton district, was reported by Allan (1940, page 220) to be thoroughly established and growing with *Juncus maritimus* var. *australiensis* in salt meadows. It is now more conspicuous away from tidal land in numerous depressions on alluvium and sand and extending along State Highway 1 to Himatangi Radio Station 3 km north of Foxton.

Spartina is still abundant—in fact, it covers more land than any of the communities of native plants—but it may have decreased since 1924. Residents say it has increased recently but the spread may be recovery after earlier destruction by floods. The active colonisation and the large bare areas suitable for *Spartina* still uncolonised lead me to this conclusion.

Near the mouth of the estuary inside the northern head there are numerous circular patches of *Spartina* growing on the wet mud. The smallest and youngest of these consist of *Spartina* alone. Others have a marked zonation. The sequence from the outer margin to the centre is: flowering plants of *Spartina*; *Spartina* non-flowering and green; *Spartina* non-flowering and brown; *Salicornia* and *Samolus*; a high central area with *Salicornia* and *Apium australe*.

Near the outlet of a creek at the motor camp there is one of the few pure stands of *Scirpus americanus*, and above the high tide level on the margin of the same creek is a patch of *Scirpus caldwellii*.

Nearby is another community of note which can best be called a salt meadow in the terminology of Cockayne (1928, page 85). However, it differs from most salt meadows in that the *Salicornia* (a distinctly reddish form) grows on raised circular patches of sand and is reached by only the highest tides or floods (figure 21). In most aspects it resembles the salt meadows of Farewell Spit of the South Island, New Zealand, more closely than any other known to me. Like the Farewell Spit communities this one receives considerable amounts of sand carried from the neighbouring dunes by wind.

At the lowest elevations, i.e., towards the river, the *Salicornia* mounds cover

Figure 21 Salt meadow in the estuary of the Manawatu River. The community at this point consists mainly of mounds of *Salicornia* and tufts of marram established from seed. This is the only vegetation of its kind in the Manawatu district.

1969



about one-quarter of the ground surface. Towards the dunes the patches are closer together and *Salicornia* is accompanied by conspicuous amounts of *Apium australe*, *Polypogon monspeliensis*, and *Samolus*. Other significant plants on and amongst the mounds are *Selliera radicans*, *Lachnagrostis filiformis*, *Deyeuxia billardieri*, *Triglochin striatum*, *Lobelia anceps*, *Agrostis stolonifera*, *Parapholis incurva*, *Cotula coronopifolia*, *Plantago coronopus*, *Scirpus cernuus*, *S. fluviatilis*, *S. nodosus*, *S. setaceus*, *Chenopodium ambiguum*, *Puccinellia fasciculata*, *Carex litorosa*, and *Juncus maritimus* var. *australiensis*.

Nearer the dune greater quantities of sand blanket the flat land almost obliterating the mounds and providing a habitat for both *Salicornia* and marram. Ultimately a dune would form here unless floods or exceptionally high tides carry away the sand and maintain the habitat in favour of *Salicornia*. In an environment where sand drifts freely and high waters periodically sweep the flat, nothing can be regarded as stable or permanent.

Streams and riparian wetlands

Slow-flowing streams

Variations in the nature of waterways bring variations in the associated floras. Only one species, *Callitriche stagnalis*, occurs in most situations where there is abundant moisture. At the one extreme the fast-flowing streams are almost devoid of higher aquatic plants in the channel or on the banks. At the other extreme streams which flow slowly have a wide range of species. The difference is brought about mainly by the quantity of moist silt in which plants can establish a foothold.

The streams flowing rapidly over shingle or solid greywacke support very few aquatics because of the scouring action of the water and shingle. The slower streams contain *Potamogeton crispus* more frequently than any other submerged species. Increasing amounts of fine sediments encourage *Elodea* and sometimes *Ottelia* but neither is abundant. In the backwaters of the Manawatu River downstream from Hamiltons bend *Myriophyllum aquaticum* is abundant in some places in spite of recent efforts to eliminate it. *Egeria densa* occurs in the Manawatu district but is not widespread at present. *Myriophyllum elatinoides*, *M. propinquum*, and *Potamogeton cheesemanii* occur in many streams but are of little significance. *Potamogeton ochreatus*, recorded in the Moutoa region by Poole and Boyce (1949), was not seen during this survey but could have been overlooked. *Potamogeton pectinatus* grows in many streams associated with the coastal lagoons.

The silty margins of sluggish streams and backwaters encourage a flora which is predominantly exotic. *Callitriche stagnalis* flourishes on the open muddy places, usually with *Rorippa sylvestris* which is widely distributed on all moist places within flood level along the Manawatu River. *Rorippa islandica* is less frequent. *Bidens frondosa* and *Galium palustre* are more common near the coast than elsewhere and are absent from apparently suitable sites inland. *Glossostigma elatinoides* is encountered in a few places. The same habitat may also be occupied by *Polygonum decipiens*, *P. hydropiper*, *Ranunculus sceleratus*, *Agrostis stolonifera*, and *Rumex conglomeratus*.

Shingly river beds

There are many species along shingly river beds in the lowlands which are not aquatics or plants of damp places but are associated with waterways for some other reason. In many instances they grow in such places because periodic flooding and the deposition of silt create an open habitat. A few others are there because of low grazing intensity. The first group includes the annuals *Chenopodium ambrosioides*, *Navarretia squarrosa*, and *Erigeron canadensis*—all species rarely found outside this habitat in the Manawatu, except where shingle has been taken from the river bed and deposited somewhere else. Another species of interest in this situation is *Aster subulatus*. Since I first saw it in about 1962 it has appeared in many places along the course of the Manawatu River. It seems to be a recent introduction which has plenty of potential to increase its area.

More than 70 species establish on the shingle domes of the Manawatu River bed when the level of the river is low. These plants of open habitats usually have short occupancy, most being washed away by the next flood or dying due to submergence. Some wetland species such as *Cyperus eragrostis* and *Agrostis stolonifera* usually maintain a hold along with *Raoulia tenui-*

caulis var. *dimorpha*, the only plant specific to this habitat. Other native plants here are several creeping *Epilobium* spp., *Glossostigma elatinoides*, *Schizeilema nitens*, *Erechtites minima*, and *Haloragis erecta*.

Each year, in late October or early November, seedlings of *Salix viminalis* appear on the moist silt of the Manawatu river bed near where both male and female parents have been planted for river bank protection. Establishment is aided by showery weather at the time of seed fall. The rate of growth varies markedly with the nature of the soil. In deep moist silt seedlings may reach 1.5 m within a year and by the following January have doubled their height. Seedlings in sand seldom reach more than 30 cm in height in 15 months.

At higher levels on the river margins there are extensive areas of tall fescue, goat's rue, tree lupin, and *Calystegia sepium* subsp. *silvatica* where floods periodically spread silt. Because intensive grazing of these accretion areas cannot be achieved this community persists.

Ditches and drains

Ditches and other small drains occasionally carrying water provide a habitat for a limited spectrum of species ranging from true aquatics to plants which normally inhabit moist places. In these waterways submerged plants are usually absent, their place being taken by *Callitriche stagnalis* which is versatile enough to flourish across the surface of the water, totally submerged or on wet mud unless shaded out by *Rumex conglomeratus*, *Scirpus fluviatilis*, *Polygonum decipiens*, *P. hydropiper*, *Cyperus eragrostis*, or *Glyceria declinata*. Where *Glyceria maxima* and *Phalaris arundinacea* thrive most other species are excluded. These rank grasses are most abundant in the lower reaches of the Manawatu River. Fortunately, most drains in the Kairanga County are fairly free of them.

Larger drains all have *Potamogeton crispus* and some have *Elodea* as well. *Myriophyllum aquaticum* and *Egeria* are mainly close to the Manawatu River. The banks of the large drains have remarkably few wetland species because of grazing close to the edge of the water and the steep gradient of the bank. In the narrow margin near the water *Polygonum* spp., *Alisma*, and *Glyceria declinata* may be found unless replaced by *Glyceria maxima* and *Phalaris arundinacea*.

Manawatu River cut-off

Near the disused wharf in the Foxton loop is an assemblage of plants which is not duplicated elsewhere in the Manawatu (figure 22). This reach no longer carries the full flow of the river, the water movement is slow, and it seems that silt has accumulated in large quantities. Although the water level rises and falls with the tides it is apparent that salt water does not regularly reach this point. *Ruppia polycarpa* and *Egeria densa* fill the channel and along



Figure 22 The Foxton loop, once the main channel of the Manawatu River but now a backwater. On the soggy margins are *Glyceria maxima*, *Alisma*, *Apium nodiflorum*, *Eleocharis acuta*, and *Phalaris arundinacea*. There are scattered clumps of raupo and *Scirpus lacustris*. The near channel is filled with *Egeria*. 1969

the soggy margins grow a luxuriance of *Eleocharis acuta*, *Alisma*, *Glyceria maxima*, *Phalaris arundinacea*, *Apium nodiflorum*, *Polygonum decipiens*, and *P. hydropiper*. There are a few patches of raupo and *Scirpus lacustris*. On the banks tall fescue and goat's rue predominate.

Hill country wetland communities

In the hill country very interesting wetland communities with many native and introduced herbaceous plants have developed where forests stood less than a century ago. The best examples are on Browns Flat, a basin at the head of the Tiritea Stream (figure 12). Here on an area of about 162 ha there are more than 32 km of streams forming a dendritic drainage pattern. After the forest was cleared it seems that *Juncus articulatus* established in the creeks and, by slowing the flow and gathering silt, created a habitat for many wetland species. This sort of habitat extends from the seepages from which the streams arise, down steep slopes and across the floor of the basin and ends abruptly where the main stream enters the bush.

In the stream beds most of the cover is provided by *Juncus articulatus* but it is regularly accompanied by *Stellaria alsine*, kneed foxtail, *Eleocharis*

acuta, *E. gracilis*, *Scirpus prolifer*, *S. sulcatus* var. *distigmatus*, *S. setaceus*, and mosses. Other species of less importance are *Epilobium rotundifolium*, *E. insulare*, *Juncus planifolius*, *Myosotis caespitosa*, *Luzula congesta*, *Gnaphalium limosum*, and *Schoenus maschalinus*. On the margins of the streams there are *Carex virgata*, *C. dissita*, *C. ovalis*, *Blechnum capense*, *Pratia angulata*, and *Juncus gregiflorus*, but there is considerable overlapping of distributions because of irregularities of topography influencing the wetness of the site. In damp level areas turf is formed by *Eleocharis gracilis*, *Juncus articulatus*, *Centella uniflora*, selfheal, *Haloragis micrantha*, *Potentilla anglica*, and creeping buttercup.

On the floor of the basin there is some *Sphagnum* moss, *Cortaderia toetoe*, *Nasturtium microphyllum*, *Epilobium pallidiflorum*, *E. chioanthum*, and raupo.

Kahuterawa Flat presents a similar picture but its flushes are very distinctive. They contain the only *Drosera binata* in the district and several species not normally associated with habitats as wet as these. The very abundant species are *Juncus articulatus*, *Eleocharis gracilis*, *Leontodon taraxacoides*, *Lotus pedunculatus*, sweet vernal, *Haloragis micrantha*, *Juncus gregiflorus*, and *Stellaria alsine*. Species which are less abundant but still a significant part of the vegetation are *Plantago hirtella*, *Hypericum japonicum*, suckling clover, *Luzula congesta*, *Ranunculus repens*, and selfheal. A sword-leaved rush, *Juncus ensifolius*, occurs in a few places.

Browns Flat and Kahuterawa Flat provide many examples of native species, formerly restricted to very limited areas, expanding their distribution following the removal of forest. They are being protected from strong competition from aliens by grazing animals. It is very apparent that if grazing animals were removed grasses would soon overwhelm the smaller herbs. Such a community Cockayne would have classified as "indigenous induced".

SHRUBLANDS

Shrubland developed where fire or other factors prevented woody vegetation running the full course of succession to forest. Shrubland occupying the wet sand plains, the river margins and the summit of Awaru could broadly be termed "natural" but the possibility of fire in each case cannot be ruled out. Other parts of the sand plains and the downlands carried shrubby vegetation which almost certainly was secondary.

Shrublands on the lowlands

Early records of scrub are very vague. Kettle's only comment (Kettle 1842)

was "manuka very prevalent about this part, particularly on the north bank" in reference to the Te Matai region upstream from Palmerston North. Stewart (S.O. 10639) coloured areas of "small bush and scrub" light green but these were not distinguishable on the monochrome prints available to me. Elsewhere scrub is indicated in only one place by the words "Scrub and toe toe" between Poplar Rd and Bowes Rd. The plan by Knowles (S.O. 10837 is an incomplete copy of it) shows "shrubby swamp" on the Makerua swamp near Linton and "scrub" on its margin nearby. On the left bank of the Manawatu River opposite the Karere Lagoon the same plan shows "koromiko scrub".

The vegetation of the plains of the second dune phase escaped mention in early records. J. W. Barber (pers. comm.) recalled extensive tracts of "miki" (*Coprosma propinqua*) on most wetlands about the coastal lagoons and extending inland just north of Foxton to State Highway 1. It remained there until cleared from most of the area 40 to 45 years ago. Its distribution seems to have coincided with the soils mapped as Pukepuke black sand. The composition of this vegetation can be judged from the remaining stands. They contain few species other than *Coprosma propinqua* and *C. propinqua* × *robusta*. The term "miki" includes *Olearia virgata* which has a wide habitat range, and according to E. Dear (pers. comm.) was abundant in swampy places. There is still some in these habitats.

According to J. W. Barber, manuka occupied the dry parts of the sand plains, apparently growing on soils mapped as Himatangi sand and Himatangi weakly mottled sand. This shrubland extended onto the dunes where manuka was associated with kanuka. Again we cannot be sure that this was the "natural" vegetation. It may have been the vegetation which developed after the plant cover was burnt by the early settlers or the Maoris. There is some manuka and bracken on the lowlands today but the area is very limited.

The shrubland on the margins of the rivers is difficult to reconstruct from the meagre accounts available. There is enough to suggest that manuka was abundant in some places, probably clothing the deposits of alluvium. A record by Knowles (original of S.O. 10837) indicates that "koromiko" (presumably *Hebe stricta*) grew on the river margins and this is in accord with what is known of its ecology. It has a habitat range from swamps to dry sunny faces and could have established at many places in the native vegetation. At present tree lupin is probably the main lowland shrub. It grows on the sand country and the shingly river beds in dry habitats.

Shrublands on the downlands

The statements about the downlands shrublands have been simple and unvarying, regardless of the source. The plants listed are manuka, toetoe

(presumably *Cortaderia toetoe*), tutu (presumably *Coriaria arborea*), koromiko (presumably *Hebe stricta*) and bracken (stated to be 3 m high). The only shrubland today is small patches of gorse and manuka.

Shrubland on Arawaru

The shrubland on Arawaru is unmatched elsewhere in the Manawatu but does not differ in any major way from other *Olearia colensoi* stands in the Tararua and Ruahine Ranges. It appears that forest of kamahi and Hall's totara gave way at around 670 m to *O. colensoi* which extended to the summit at 767 m. Now, pepper tree, *Cyathea smithii* and a few other species of low stature have replaced the forest and the timber line is undefined. The *Olearia* stand is in a state of flux, about one-half having died out progressively from the margins inwards. These are being invaded by *Coprosma* cf. *parviflora*, *Phormium cookianum*, *Chionochloa conspicua*, *Paesia scaberula*, and *Histiopteris incisa*. The decadence of the *Olearia* appears catastrophic but the same species is returning. This is particularly clear on the eastern side of the summit where a new crop is about 0.6 m high and sharing the canopy with *Coprosma*. There is no shortage of *Olearia* seedlings anywhere. Perhaps this is a normal mode of rejuvenation of stands.

The associated flora is of particular interest as at least 11 species do not occur in any other part of the Manawatu. In this category are placed the following plants—*Senecio elaeagnifolius*, *Olearia lacunosa* (only one very old specimen seen), *Gaultheria depressa*, *Gentiana mathewsii*, *Chionochloa rubra* × *flavescens*, *Astelia nervosa*, *Uncinia filiformis*, *U. caespitosa*, *Scirpus habrus*, *Aporostylis bifolia*, and *Hymenophyllum armstrongii* hidden away among mosses on the stems of *Olearia colensoi*. Other species noted in the *Olearia colensoi* community include *Coprosma* sp., *Gaultheria antipoda*, *Myrsine divaricata*, *Dracophyllum filifolium*, *Helichrysum bellidioides*, *Epilobium pedunculare*, *E. pedunculare* var. *brunnescens*, *E. alsinoides*, *Phormium cookianum*, *Gabnia* sp., *Carex dissita*, *Uncinia ferruginea*, *U. rupestris*, *Chionochloa conspicua*, *Polystichum vestitum*, *Paesia scaberula*, *Histiopteris incisa*, *Blechnum* sp., *Lycopodium scariosum*, *L. volubile*, and *L. fastigiatum*.

Some additional species were noted by A. P. Druce approaching Arawaru from the eastern side in November 1959. Some were possibly within the boundaries of the Manawatu. These are of particular interest because they were not recorded by me as plants of the Manawatu, because they were not found or were overlooked. They are *Cyathea colensoi*, *Hymenophyllum lyallii*, *Haloragis depressa*, and *Euphrasia cuneata*.

Present-day shrublands of the water reserves

Except for the tawa forest, the plant cover of the water reserves in the Tiritea and Kahuterawa valleys bears little resemblance to the forest it has replaced in the last few decades. Most of the non-agricultural land has degraded to shrubland and clearings. In some places scraps of forest with emergent rimu and miro persist while in others turf only a few centimetres high grows where forest trees more than 33 m tall grew a century ago (figure 19). Between the two extremes is shrubland composed principally of mahoe, pigeonwood, *Pseudowintera* spp., and *Cyathea smithii*. In this section attention is focused on the Tiritea region but the remarks apply to the upper parts of the Kahuterawa and Otangaue catchments which still carry similar native vegetation. The changes in the vegetation of the Tiritea reserve were the subject of a special study (Esler 1969c).

Destruction of the original forest cover is attributed to fire, opossums, deer, pigs, and cattle but other factors have played a secondary role at various stages of the degeneration sequence (figure 23).

There is some evidence that fire swept part of the Tiritea catchment about two centuries ago but the general deterioration of all the forest has obliterated the boundaries of the burn. There is ample evidence that at the time adjoining land was being cleared for farming, fires reached into the reserve in many places. These areas are marked now by an abundance of rangiora.

Without doubt, opossums were primarily responsible for the death of the kamahi and it is probable that they played a part in the destruction of the northern rata also. The first record of liberation of opossums in the Tiritea

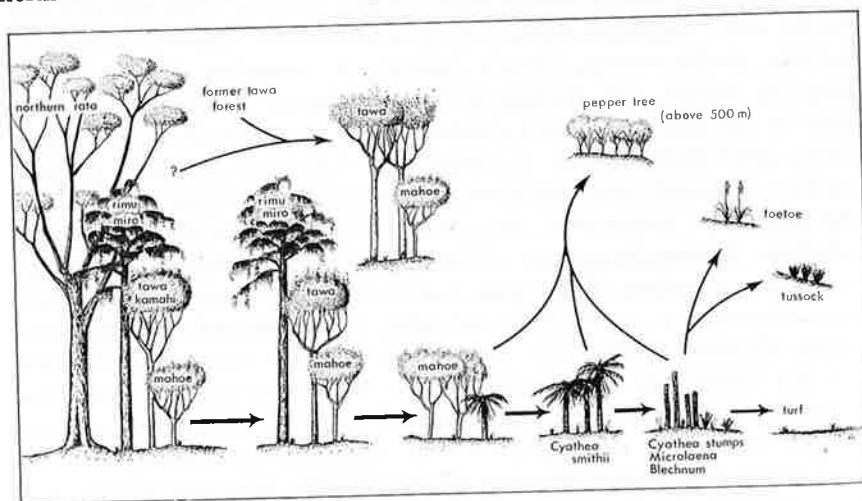


Figure 23 The course of botanical changes in the Tiritea catchment.

given by Pracy (1962) is 1919 but F. G. Hayes, who has kindly supplied me with a great deal of useful information about the catchment, stated that "Possums were fairly numerous about the caretaker's house in 1918" and that the main liberation was in 1907. Opossums were numerous when the first season opened in 1921. Large numbers were taken that year and for many years after. The population which built up from these liberations may have been reinforced by migrants from Shannon, less than 24 km away, where the Wellington Acclimatisation Society made a liberation in 1892.

Because hunters and trappers failed to notice the demise of the dominant trees, kamahi and northern rata, we have no record of when the destruction took place. Hayes stated that in 1927 the forest was open and the floor was clean but there was a good canopy overhead. V. D. Zotov (pers. comm.) believes that the drastic changes in the vegetation occurred after 1940. The 1949 aerial photographs show considerable canopy deterioration. At least the beginnings of all the major clearings are visible.

The opening of the canopy would normally have stimulated a flush of vegetation but removal of seedlings by ground browsers allowed limited regeneration—only that which out-paced the browsers' capacity to destroy it.

Figure 24 Forest deterioration on the humid sheltered margins of the Tiritea Stream is masked by the abundance of tree ferns, mostly mamaku. 1964





Figure 25 *Griselinia littoralis*, *Coprosma australis* and *C. foetidissima* epiphytic on tree ferns above the reach of deer. In the absence of deer these plants would grow on the ground and assist the return of forest. 1964

Mahoe, pigeonwood, toro, and heketara formed a new short-lived canopy. Death of toro and mahoe, seemingly accelerated by opossums, left gaps in the canopy that were filled by *Cyathea smithii* which persisted only as long as shrubs remained to give protection from sun and wind.

Exposure to strong, persistent, salt-laden Manawatu winds has hastened the deterioration once the canopy has broken but the same process of decline is also operating in sheltered sites (figure 24). This highlights the one major factor effective in all places—the failure of seedlings of most trees and shrubs to reach maturity. The only place where plants palatable to deer can develop is above the browse level, usually on tree ferns which are short-lived (figure 25).

When *Cyathea smithii* wanes, *Uncinia uncinata*, *Microlaena avenacea*, and *Blechnum discolor* flourish, but they, too, are dependent on shade and shelter for survival. As the stumps of shrubs and tree fern disintegrate only turf and tussock survive.

The rise to dominance of pepper tree in places above 500 m probably began with the destruction of the primary forest but has been hastened by the decadence of the secondary forest. In the parts exposed to the full blast of the winds pepper tree seems to have been affected by salt damage. In these places very dense stands of *Pseudopanax anomalus* have developed.

The present-day cover of the Tiritea and Kahuterawa catchments is a mosaic, each phase a stage in the degradation sequence. In some places the sequence is regular and orderly, but some tenacious plants have withstood the changes while others around them have succumbed. Thus we sometimes find rimu or miro trees surrounded by turf reminding us that, but for introduced animals, the environment is still capable of supporting trees. Fences erected in the Tiritea reserve to protect sample areas of vegetation from browsing animals have made it very clear that forest has the capacity to return.

Other present-day shrublands

Most other shrubland is in places where intensive grazing is not possible because of the steepness of the land or the infertile soil. On the steep greywacke country native shrubs are returning in many places following bracken. On the terraces manuka occupies large areas of low natural fertility; three other types of shrubland are dominated by gorse, pepper tree, and tree lupin respectively.

Regeneration through bracken

The greywacke slopes near streams (figure 26) have some good examples of regeneration following the clearing of the forest for farming. The sequence is arrayed in space as well as time because the return of native plants has begun near the creeks and progressed upslope. The stages can be traced in a transect from the pasture to the creek but it must be realised that near the creek the vegetation is not only older but the environment is also more favourable for forest. Also, burning may not have been so intense near the streams. The succession begins as bracken moves into the margin of the pasture. Sheep and cattle continue to penetrate as long as some pasture remains. Before animals are excluded by the invading bracken, seedlings of native shrubs have already begun to appear under the fern canopy. Paradoxically, regeneration is much more rapid here than in continuously ungrazed areas. In the latter situation absence of grazing animals leads to tall dense bracken which occupies a site almost indefinitely and permits few seedlings on the densely shaded ground strewn with fronds from previous seasons. Where animals have access the fronds are shorter and the debris does not accumulate so freely, but there is sufficient protection for seedlings to establish. By the time the seedlings have reached above the bracken the area generally has little attraction to the sheep and cattle. Ardent conservationists seldom realise that in many situations poor farming with sheep will hasten regeneration much faster than fencing will encourage it.

The early entrants into the fern stands are putaputaweta, kaikomako, mahoe,